

PhoneZone — Mobile Shop App

CS2430 Mobile Computing | Exercise 1 | Kotlin Jetpack Compose

Overview

PhoneZone is a phone shopping app built with Kotlin and Jetpack Compose. It lets users browse phones, view specs, pick a color and storage option, add items to cart, and place an order. All product data is stored in a plain Kotlin list — no database is used.

Project Structure

File	What it does
MainActivity.kt	Entry point. Sets the Compose theme and launches the nav graph.
data/Product.kt	Product data class with all phone fields.
data/CartItem.kt	Holds a product, chosen color, storage, and quantity.
data/ProductData.kt	List of 6 phones (Samsung, iPhone, Pixel, OnePlus, Xiaomi).
viewmodel/ShopViewModel.kt	Manages cart state, filtering, and totals.
ui/navigation/NavGraph.kt	4 routes: home, detail/{id}, cart, success.
ui/screens/HomeScreen.kt	Product grid, search bar, category filter.
ui/screens/ProductDetailScreen.kt	Phone detail, color/storage picker, add to cart.
ui/screens/CartScreen.kt	Cart list, quantity controls, order total.
ui/screens/SuccessScreen.kt	Order confirmation screen.
ui/theme/	App colors and typography.

Requirements

- Android Studio Hedgehog or newer
- JDK 21
- Android SDK 35 (minSdk 24, targetSdk 35)
- Internet access on first Gradle sync to download dependencies

How to Run

1. Open the project	In Android Studio or IntelliJ: File → Open → select the MobileShop folder.
2. Sync Gradle	Click Sync Now when prompted, or File → Sync Project with Gradle Files.
3. Pick a device	Choose an emulator (e.g. Pixel 6, API 34) from the toolbar dropdown.
4. Run	Press the green Run button. The app opens on the PhoneZone home screen.

Physical device	Enable Developer Options on the phone, turn on USB Debugging, plug in, select from device list.
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Key Dependencies

- Jetpack Compose BOM 2024.12.01 — all Compose libraries
- material-icons-extended — full Material icon set (needed for Remove icon)
- navigation-compose 2.8.5 — screen navigation
- coil-compose 2.7.0 — async image loading
- lifecycle-viewmodel-compose — ViewModel in Compose

Gradle & Java Versions

Tool	Version
Gradle Wrapper	8.11.1
AGP (Android Gradle Plugin)	from settings.gradle
Java / JVM target	21
compileSdk / targetSdk	35
minSdk	24

GitHub

Upload the project to a public GitHub repo: `git init` → `git add .` → `git commit -m 'initial commit'` → `git push`.
Share the link with the instructor.